30 Programming Questions Knowledge Assessment Test

For Testing Programming Students | Created: July 11, 2026

EASY LEVEL - Basic Concepts

1. Check if a Number is Prime

Write a program to check whether a given number is prime or not.

- Input: 17
- Output: True (it's prime)
- Hint: A prime number is only divisible by 1 and itself

2. Count Vowels in a String

Write a program to count the number of vowels (a, e, i, o, u) in a given string.

- Input: 'Hello World'
- Output: 3 (e, o, o)

3. Reverse a String

Write a program to reverse a given string without using built-in reverse functions.

- Input: 'hello'
- Output: 'olleh'

4. Find the Largest Number in an Array

Write a program to find the largest number in an array without using built-in max() function.

- Input: [5, 2, 9, 1, 7]
- Output: 9

5. Check if a Number is Palindrome

Write a program to check if a number reads the same forwards and backwards.

- Input: 121
- Output: True
- Input: 123
- Output: False

6. Count Occurrences of a Character

Write a program to find how many times a character appears in a string.

- Input: String = 'programming', Character = 'g'
- Output: 2

7. Sum of All Numbers in an Array

Write a program to calculate the sum of all numbers in an array without using built-in sum() function.

- Input: [1, 2, 3, 4, 5]
- Output: 15

8. Check if a String is Empty

Write a program to check if a string is empty or not.

- Input: ''
- Output: True
- Input: 'hello'
- Output: False

9. Multiply Two Numbers Using Addition

Write a program to multiply two numbers using only addition (no * operator).

- Input: 5, 3
- Output: 15 (by adding 5 three times: 5 + 5 + 5)

10. Find the Minimum Number in an Array

Write a program to find the smallest number without using built-in min() function.

- Input: [15, 3, 8, 1, 22]
- Output: 1

MEDIUM LEVEL - Intermediate Concepts

11. Sort an Array Without Using Built-in Sort

Write a program to sort an array in ascending order using bubble sort, selection sort, or insertion sort.

• Input: [64, 34, 25, 12, 22, 11, 90]

• Output: [11, 12, 22, 25, 34, 64, 90]

12. Remove Duplicates from an Array

Write a program to remove duplicate values from an array.

• Input: [1, 2, 2, 3, 4, 4, 5]

• Output: [1, 2, 3, 4, 5]

13. Find the Second Largest Number

Write a program to find the second largest number in an array.

• Input: [10, 5, 20, 8, 15]

• Output: 15

14. Check if Two Strings are Anagrams

Write a program to check if two strings are anagrams (contain the same characters in different order).

• Input: 'listen', 'silent'

• Output: True

15. Reverse an Array

Write a program to reverse an array without using built-in reverse functions.

• Input: [1, 2, 3, 4, 5]

• Output: [5, 4, 3, 2, 1]

16. Find Common Elements in Two Arrays

Write a program to find elements that exist in both arrays.

• Input: [1, 2, 3, 4], [3, 4, 5, 6]

• Output: [3, 4]

17. Check if Array is Sorted

Write a program to check if an array is sorted in ascending order.

• Input: [1, 2, 3, 5, 7]

• Output: True

• Input: [1, 3, 2, 5, 7]

• Output: False

18. Fibonacci Sequence

Write a program to generate the first N numbers in the Fibonacci sequence.

• Input: 7

• Output: [0, 1, 1, 2, 3, 5, 8]

19. Find Factorial of a Number

Write a program to calculate the factorial of a given number.

• Input: 5

- Output: 120 (5 × 4 × 3 × 2 × 1)

20. Count Words in a String

Write a program to count the number of words in a string.

- Input: 'Hello World This is Python'
- Output: 5

21. Check if a String Contains Another String

Write a program to check if one string is a substring of another (without using built-in contains/in functions).

- Input: 'Hello World', 'World'
- Output: True

22. Rotate an Array

Write a program to rotate an array by N positions.

- Input: [1, 2, 3, 4, 5], rotate by 2
- Output: [4, 5, 1, 2, 3]

23. Find Most Frequent Element

Write a program to find the element that appears most frequently in an array.

- Input: [1, 1, 1, 2, 2, 3]
- Output: 1

24. Check if a Number is Armstrong Number

An Armstrong number is a number that equals the sum of its own digits raised to the power of the number of digits.

- Input: 153
- Output: True ($1^3 + 5^3 + 3^3 = 153$)

25. Convert String to Uppercase and Lowercase

Write a program to convert a string to uppercase and lowercase without using built-in methods.

- Input: 'Hello World'
- Output: 'HELLO WORLD' (uppercase), 'hello world' (lowercase)

HARD LEVEL - Advanced Concepts

26. Find Longest Common Subsequence

Write a program to find the longest subsequence that is common to two strings.

- Input: 'ABCDGH', 'AEDFHR'
- Output: 'ADH' (length: 3)

27. Two Sum Problem

Given an array and a target sum, find two numbers that add up to the target.

- Input: [2, 7, 11, 15], target = 9
- Output: [2, 7] (indices: [0, 1])

28. Merge Two Sorted Arrays

Write a program to merge two sorted arrays into a single sorted array.

- Input: [1, 3, 5], [2, 4, 6]
- Output: [1, 2, 3, 4, 5, 6]

29. Find All Permutations of a String

Write a program to generate all possible permutations of a string.

- Input: 'ABC'
- Output: ['ABC', 'ACB', 'BAC', 'BCA', 'CAB', 'CBA']

30. Check if Array is a Valid Mountain Array

An array is a valid mountain if it increases up to a peak and then decreases.

- Input: [0, 2, 3, 4, 5, 2, 1, 0]
- Output: True
- Input: [0, 1, 2, 1]
- Output: False (should go up then down, not start from 0)